WORKED SOLUTION

STEP III 2005, Question 9

Mechanics • Collisions and Restitution

STEP III 2005, Question 9

Read the question before you go on. It is in the STEP Support Programme's question database:

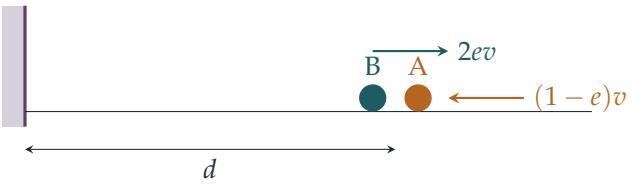
step.maths.org/questions/05-s3-q9

The key idea

The masses look strange, but they have been chosen so that the first collision gives very tidy answers. Once you have the second collision, compare the velocities just before it with the velocities just before the first one: they are the same, but each multiplied by e. Collision laws are linear in the velocities, so the whole story then repeats on a smaller scale, and that gives every later collision at once.

Solution

We model A and B as particles on a smooth horizontal line, with Newton's experimental law at each impact. Take the direction *towards the wall* as positive, and measure distances from the wall.



(The picture shows the instant just before the first impact, with the two particles drawn slightly apart.)

The first collision

Let A and B have velocities v_A and v_B (towards the wall) afterwards. Before, A has velocity $(1-e)v$ and B has velocity $-2ev$.

Conservation of momentum gives

$$\begin{aligned} 4em\,v_A + (1-e)^2 m\,v_B &= 4em(1-e)v - (1-e)^2 m \cdot 2ev \\ &= 2e(1-e)v[2-(1-e)]m = 2e(1-e)(1+e)v\,m. \end{aligned}$$

The speed of approach is $(1-e)v + 2ev = (1+e)v$, so Newton's law gives

$$v_B - v_A = e(1+e)v.$$

Substituting $v_B = v_A + e(1+e)v$ into the momentum equation and dividing by m:

$$[4e + (1-e)^2]v_A = 2e(1-e)(1+e)v - (1-e)^2 e(1+e)v = e(1-e)(1+e)v[2-(1-e)].$$

Since $4e + (1-e)^2 = (1+e)^2$, the right-hand side is $e(1-e)(1+e)^2 v$, and so

$$v_A = e(1-e)v, \qquad v_B = e(1-e)v + e(1+e)v = 2ev.$$

Both are positive: after the collision **A moves towards the wall with speed** $e(1-e)v$ **and B moves towards the wall with speed** $2ev$.

The second collision

B is faster, so it pulls ahead of A and reaches the wall after time $t_1 = d/(2ev)$. In that time A covers $e(1-e)v \cdot \frac{d}{2ev} = \frac{1}{2}(1-e)d$, so A is at distance

$$d - \tfrac{1}{2}(1-e)d = \tfrac{1}{2}(1+e)d$$

from the wall. B rebounds with speed $e \cdot 2ev = 2e^2 v$, now moving away from the wall, while A is still moving towards it with speed $e(1-e)v$. They are approaching each other, with closing speed $e(1-e)v + 2e^2 v = e(1+e)v > 0$, so they must meet again (before A could reach the wall, since B lies between them). This takes a time

$$t_2 = \frac{\frac{1}{2}(1+e)d}{e(1+e)v} = \frac{d}{2ev},$$

during which B travels $2e^2 v \cdot \frac{d}{2ev} = ed$ from the wall. So **a second collision takes place, at distance** de **from the wall**, as required.

Further collisions

> **Ask yourself** How can we deduce that there are more collisions without doing all the work again?

Look at the state just before the second collision: A moves towards the wall with speed $e(1-e)v$ and B moves away from it with speed $2e^2 v = 2e(ev)$. This is exactly the initial state with v replaced by ev and d replaced by de (the masses and e are unchanged). Everything we did above holds for any $v > 0$ and $d > 0$, so the same argument applies with (ev, de) in place of (v, d): there is a third collision, at distance $(de)e = de^2$, and just before it the state is the initial one with $(e^2 v, de^2)$. By induction there is an $(n+1)$th collision for every n, so collisions go on for ever.

Just before the nth collision the state is the initial one with v and d replaced by $e^{n-1}v$ and de^{n-1}, so

the nth collision takes place at distance de^{n-1} from the wall.

From the second part, the time from one collision to the next is $t_1 + t_2 = d/(ev)$. Starting from the nth collision, this becomes

$$\frac{de^{n-1}}{e \cdot e^{n-1}v} = \frac{d}{ev},$$

which does not depend on n. So the times between successive collisions are all equal, to $d/(ev)$. The distance and the speeds both shrink by the factor e each cycle, and these two effects cancel in the time.

Checks

- Momentum in the first collision: with $e = \frac{1}{2}$, $m = v = 1$, the masses are 2 and $\frac{1}{4}$; before, $2 \cdot \frac{1}{2} - \frac{1}{4} \cdot 1 = \frac{3}{4}$; after, $2 \cdot \frac{1}{4} + \frac{1}{4} \cdot 1 = \frac{3}{4}$. The relative speed goes from $\frac{3}{2}$ to $\frac{3}{4}$, a factor e.
- We simulated the motion exactly (with fractions) for 200 random choices of e, v, d, m and eight collisions each: every collision was at de^{n-1} and the gaps were all $d/(ev)$. For $e = \frac{1}{2}$, $v = d = 1$ the collisions are at times $0, 2, 4, 6$ and distances $1, \frac{1}{2}, \frac{1}{4}, \frac{1}{8}$.

Common slips

- Mixing up the signs: B starts out moving *away* from the wall, so its initial velocity is negative in our convention.
- Forgetting that the wall also has coefficient e, so B leaves the wall with speed $2e^2v$, not $2ev$.
- Redoing the second collision by brute force instead of noticing the scaling; it works, but it's much longer and the pattern is easy to miss.

Beyond STEP

The deduction here uses linearity and *self-similarity*: the equations of the collisions are unchanged if every velocity is multiplied by the same constant, so a state that reproduces itself up to scale repeats for ever. The same idea underlies the analysis of a bouncing ball with infinitely many bounces, and dimensional analysis more generally. Kleppner and Kolenkow, *An Introduction to Mechanics*, treats collisions and momentum with this kind of care.

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